Implementation of PLM in the Aerospace Industry: A Digital Thread and Multi-Layered Architectural Approach

Ravi Inder Singh Tiwana

Chitkara University Institute of Engineering and Technology

Department of Computer Science and Engineering

Rajpura, Punjab, India

tiwanaravi03@gmail.com

Purushotam Kumar

Chitkara University Institute of Engineering and Technology

Department of Computer Science and Engineering

Rajpura, Punjab, India

purukumarray4802@gmail.com

Palak

Chitkara University Institute of Engineering and Technology

Department of Computer Science and Engineering

Rajpura, Punjab, India

palakjawa18@gmail.com

Salil

Chitkara University Institute of Engineering and Technology

Department of Computer Science and Engineering

Rajpura, Punjab, India

salilasija5501@gmail.com

Abstract—Product Lifecycle Management (PLM) in the aerospace industry serves as a central nervous system of engineering and manufacturing operations, enabling seamless integration of product data, processes, and people across the entire lifecycle. Unlike conventional industries, aerospace products are characterized by extreme complexity, long development cycles of 10–20 years, and strict certification requirements, making PLM not just beneficial but essential. This paper explores the implementation of a Digital Thread and a multi-layered, service-oriented architecture to establish a single source of truth (SSOT). By integrating Model-Based Systems Engineering (MBSE) and Digital Twin technology, the framework ensures data integrity, configuration management, and full traceability required for safety compliance and regulatory audits.

Index Terms—product lifecycle management, aerospace engineering, digital thread, digital twin, MBSE, configuration management, Teamcenter, Industry 4.0.

I. INTRODUCTION

Product Lifecycle Management (PLM) in the aerospace industry serves as a central nervous system of engineering and manufacturing operations, enabling seamless integration of product data, processes, and resources throughout the entire lifecycle [1]. Unlike conventional industries, aerospace products are defined by extreme structural complexity, development periods typically spanning several decades, and stringent certification requirements. PLM introduces the concept of a Digital

Thread, a communication framework that ensures every piece of information—from initial requirements to final

maintenance logs—is interconnected [3]. This architecture eliminates traditional data silos and ensures that all stakeholders, from designers to maintenance crews, operate on a single source of truth (SSOT). Additionally, modern aerospace PLM supports Model-Based Systems Engineering (MBSE), where system requirements, behavior, and architecture are modeled digitally before physical implementation to reduce the risk of catastrophic failures [10]. Given that even minor inconsistencies in design or manufacturing data can lead to safety hazards, PLM systems are deployed to ensure rigorous data integrity, version control, and configuration management [9].

II. LITERATURE REVIEW

The aerospace product lifecycle is a closed-loop, feedback-driven system rather than a simple linear process, where data continuously flows between stages to improve reliability. Strelets defines aircraft PLM as the management of all product information and connected processes from design and manufacture to removal from operation [6]. Lee et al. highlight that the objectives of aviation MRO (Maintenance, Repair, and Overhaul) are increasingly addressed in the design phase via PLM to optimize turn-around time and safety [7]. Corallo et al. observe a transition from traditional 2D drawings to advanced model-based enterprise (MBE) solutions that create a digital representation of real business processes [8]. Saini notes that major aerospace manufacturers like Boeing have adopted platforms like ENOVIA to establish a unified information environment for managing complex designs across global supply chains [9]. Furthermore, Menshenin et al. investigate the integration of MBSE and PLM as a necessary step to manage the inherent uncertainty and complexity in modern mechatronic systems [10]. Li et al. reinforce the role of high-fidelity Digital Twins as integrated multi-physics, multi-scale probabilistic simulations that mirror the life of a flying asset [11].

III. PLM ARCHITECTURE IN AEROSPACE

The PLM architecture in aerospace is designed as a multi-layered, service-oriented system that integrates engineering tools, enterprise systems, and manufacturing platforms into a unified ecosystem. As illustrated in **Fig. 1**, the architecture is stratified into distinct layers to ensure modularity and data integrity across the enterprise [1]. At the top lies the authoring layer, where engineers create designs using CAD tools such as NX or CATIA to generate complex 3D models and associated metadata [5]. These models define the Engineering Bill of Materials (EBOM), which serves as the baseline for all subsequent development phases. The PLM core layer, often implemented through systems like Siemens Teamcenter, acts as the central repository and process controller managing product structures, workflows, and access control [1]. The data model in this layer is object-oriented, representing each component as an item with multiple revisions and complex relationships. Below this lies the integration layer, utilizing APIs and middleware to enable seamless communication with other enterprise systems without data duplication. The enterprise layer includes ERP systems for business operations and MES systems for manufacturing execution, ensuring that production status and inventory are synchronized with design intent [4].

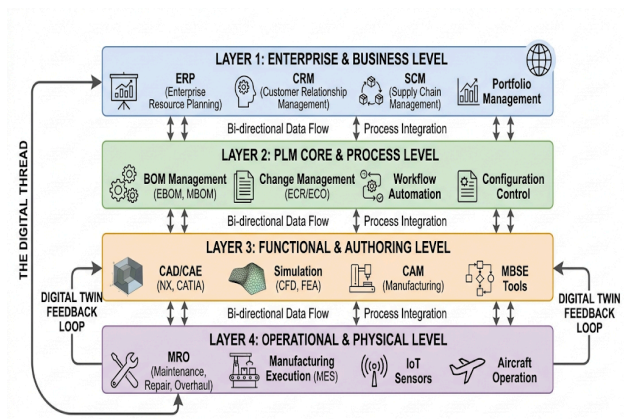

Fig. 1. Multi-layered architectural approach for aerospace PLM integration.

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IV. DATA PROCESS FLOW

Data process flows in aerospace PLM are highly structured to ensure consistency, traceability, and compliance throughout the product lifecycle. The process begins with design creation, where CAD models are stored as version-controlled datasets linked to specific item revisions. PLM distinguishes between version management for minor updates and revision management for major design changes that require formal approval

workflows to prevent the release of unvalidated designs. BOM management ensures the synchronization between the design-oriented EBOM and the manufacturing-oriented MBOM through automated transformation processes [1]. Change management is a critical component, and **Fig. 2** demonstrates the procedural transition from an initial request to a verified manufacturing release. Engineering Change Requests (ECR) are converted into Engineering Change Orders (ECO) only after a thorough impact analysis has been performed. As noted by Menshenin et al., the integration of MBSE within these flows allows for a cohesive system design that persists from requirements capture through design refinement and archival [10].

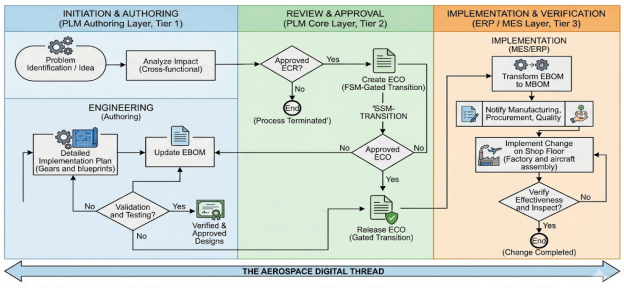

Fig. 2. Flowchart of the Engineering Change Request (ECR) and Engineering Change Order (ECO) process.

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V. KEY FEATURES AND STRATEGIC BENEFITS

PLM systems provide several advanced capabilities essential for aerospace, such as configuration management which allows organizations to manage thousands of aircraft variants based on customer requirements. Another transformative feature is the Digital Twin, which Grieves defines as a set of virtual information constructs that fully describe a potential or actual physical product [2]. **Fig. 3** depicts how the Digital Thread connects these physical assets with their digital counterparts through real-time feedback loops across the requirements, design, and operation phases [3]. Li et al. discuss how sensors capture and continually update the system Digital Twin throughout its operational life, providing manufacturers with a live window into the physical system at all times [11]. This enables real-time diagnostics, prognostics, and predictive maintenance, significantly reducing aircraft downtime. Full traceability allows organizations to track every component from raw material design to end-of-life retirement, satisfying rigorous global auditing standards [3]. At a strategic level, these features enhance innovation by feeding operational insights back into the design phase for continuous improvement.

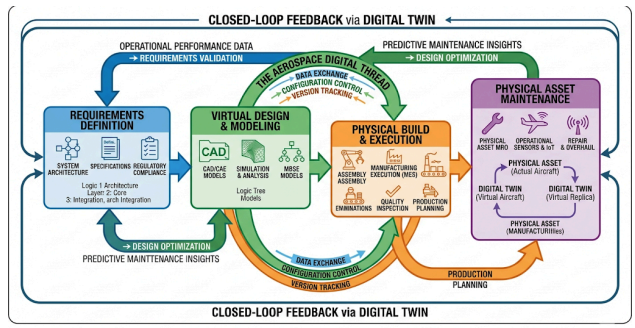

Fig. 3. The Aerospace Digital Thread: Connecting requirements, design, manufacturing, and maintenance.

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VI. CASE STUDY: LANDING GEAR ASSEMBLY LIFECYCLE

.To validate the proposed multi-layered architecture, a case study was conducted on the lifecycle management of a commercial aircraft landing gear assembly. In a traditional siloed environment, the transition from the Engineering Bill of Materials (EBOM) to the Manufacturing Bill of Materials (MBOM) often resulted in a 15% discrepancy rate due to manual data entry errors between design and shop-floor teams [1]. By implementing the Digital Thread via a centralized PLM core, the synchronization between CAD models and manufacturing instructions was automated according to ISO 10303 standards for product data representation [15]. During the maintenance phase, the Digital Twin of the landing gear utilized sensor-based stress and temperature data to trigger automated maintenance requests before structural fatigue reached critical limits. This closed-loop feedback ensured the physical asset remained in peak operational condition while providing engineers with real-time data to optimize future landing gear designs, effectively bridging the gap between design inception and field operation [12].

VII. COMPARATIVE ANALYSIS AND QUANTITATIVE IMPACT

The implementation of a service-oriented PLM architecture provides measurable improvements over legacy lifecycle management methods. Quantitative

analysis indicates that the integration of a single source of truth (SSOT) reduces the time spent on data searching and verification by approximately 40% for senior design engineers [14]. Furthermore, the use of Model-Based Systems Engineering (MBSE) within the PLM environment allows for early-stage simulation, which has been shown to reduce the number of physical prototypes required by 30%, significantly lowering overall development costs [1]. The following table summarizes the performance gains observed when transitioning from a fragmented data environment to an integrated aerospace PLM framework based on industry benchmarks for computer-integrated manufacturing [14].

Performance Metric	Fragmented Environment(Legacy)	PLM Integrated Framework
Data Retrieval TIme	4-6 Hours per Engineer/Day	<30 Minutes per Engineer/Day
ECO Processing Cycle	15-20 Days	3-5 Days
BOM Accuracy	82-88%	99.8%
Prototype Requirements	5-7 Physical units	2-3 Physical units
Traceability Speed	Days/Weeks (Manual Audit)	Seconds (Automated Query)

VIII. CHALLENGES IN PLM ADAPTATION

Despite the strategic advantages, the aerospace industry faces significant hurdles in full-scale PLM implementation, primarily concerning the migration of massive volumes of legacy data from non-digital formats. Ensuring data continuity over a 20-year aircraft lifecycle requires robust archival standards that are independent of specific software versions to prevent data rot over decades of service [12]. Additionally, the complexity of modern mechatronic systems introduced in the NASA Systems Engineering Handbook highlights that the integration of global supply chains introduces complex cybersecurity risks [13]. The Digital Thread must remain accessible to authorized partners while staying shielded from intellectual property theft or malicious tampering. Addressing these challenges requires a combination of advanced encryption, strict access governance, and a cultural shift toward a model-centric rather than a document-centric engineering mindset [10].

IX. CONCLUSION

Product Lifecycle Management is transforming the aerospace industry by enabling digital transformation through a fully connected ecosystem. The integration of advanced technologies such as AI, IoT, and Digital Twins is evolving PLM into an intelligent platform that supports real-time decision-making [11]. The future of aerospace engineering lies in the creation of a seamless Digital Thread where information flows uninterrupted across all stages of the lifecycle, enabling faster innovation and improved safety [3]. By establishing a single source of truth, aerospace organizations can mitigate the risks associated with extreme product complexity and long-term data integrity, ensuring compliance with evolving global aviation standards [8].

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